Study of Particles Separation in The Ultrasonic Microdevice

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***Abstract*—In this paper, a new MEMS device that can separate microparticles on-chip by the use of bulk-mode excited membrane vibration is proposed, designed and microfabricated. By starting from the one-dimensional analytical mode of forces acting on particles,the force for ultrasonically separating particles is profoundly studied, and finite element method in ANSYS is used to analyze the vibration modes, the acoustic field and flow field in the microdevice due to the membrane vibration. With these analyses, we obtain the ultrasonic radiation force acting on different kinds of particles in the water, blood plasma and milk medium when the microdevice vibrates at different frequencies and then study their separation.**

Keywords- particles separation; ultrasonic method; MEMS; bulk-mode excitationt

I. INTRODUCTION

Along with the notable development of microfabrication technologies, there is a fast growing demand to develop the micro-device or Lab-on-a-Chip that can fast identify, test, and analyze various kinds of bioparticles, such as bio-molecules and cells. As the first step, particles should be separated. Therefore, an effective and uncontaminating separation method is needed. At present, two types of methods, contact and non-contact methods can be used for particle separation. Since they can operate continuously and preserve the particles of interest in suspension for subsequent use without fouling the separation equipment, non-contact methods are more suitable for particle separation. Compared with other non-contact methods for separation, such as the electrophoresis method and magnetic method, the ultrasonic method only requires that the density or compressibility of the particles is different from that of the surrounding medium, it is therefore ultrasonic separation technique can also be used to separate those which are not charged particles, or nonmagnetic particles. Due to rapid development of MEMS technology over the past 20 years, investigators are focusing on the miniaturization of various types of devices or instruments with the aid of MEMS technology. In this paper, we propose and investigate a MEMS based ultrasonic device for on-chip separation of micro and/or nano particles.

II. FORCES ON PARTICLES IN ULTRASONIC FIELD

For particles suspending in a liquid, they are exposed to a stationary standing wave field and experience pressure fluctuations. This acoustic pressure gives rise to acoustic force.

According to Yosioka[1], the ultrasonic radiation force Fac on a compressible particle in the stationary standing wave field is proportional to the radius of the particle, a, the wave frequency *f*, the ultrasound energy density *E*, and the density-compressibility constant $P(\lambda,\sigma)$, which can be expressed as

$$F_{ac} = -\frac{8\pi^2 f}{c} a^3 E P(\lambda,\sigma) \sin\left(\frac{4\pi f}{c} h\right) \tag{1}$$

where *h* is the distance from the pressure node, *E* and $P(\lambda,\sigma)$ are given by

$$E = \rho_0 \frac{4\pi^2 f^2}{c^2} A^2 \tag{2}$$

$$P(\lambda,\sigma) = \frac{\lambda + \left[2(\lambda-1)/3\right]}{1+2\lambda} - \frac{1}{3}\sigma \tag{3}$$

where λ and σ are

$$\lambda = \rho^* / \rho_0 \tag{4}$$

$$\sigma = r^* / r_0 \tag{5}$$

where ρ^*, ρ_0, r^* and r_0 are the density and compressibility of particle and medium, respectively.

In eq. (1), if $P(\lambda, \sigma)$ is positive, the acoustic radiation force will make particles to be concentrated at the nodes, if $P(\lambda, \sigma)$ is negative, the acoustic radiation force will make particles to be concentrated at the antinodes. For most particles suspending in aqueous solution, such as biological cells and molecules, $P(\lambda, \sigma)$ is positive, so they move towards nodal plane [2]. This specialty of ultrasound standing wave field can be used to separate or manipulate micro and/or nano particles.

For low Reynold's number (Re<0.2), the fluid flow is laminar flow. According to Stokes Law, the drag force on a particle is calculated as

$$F_{vis} = 6\pi\eta a v \tag{6}$$

where η is the coefficient of the viscosity of the medium, a and v are the radius and velocity of particle respectively. This expression is suitable for pure water, blood plasma and milk.

The resultant force of the gravity and buoyancy is determined by the relative densities of the particle and fluid, which can be described as

$$F_g = \frac{4\pi}{3}(\rho^* - \rho_0)a^3 \tag{7}$$

When particles are separated in the ultrasonic standing wave field, they are affected under the forces already discussed. We define the resultant force of these forces as separation force, which can be described as

$$F_{sp} = F_{ac} - F_g - F_{vis} \tag{8}$$

III. Design and Fabrication of Microdevice

We propose a new MEMS ultrasonic separation microdevice, shown in Fig. 1. It is based on the principle of flexural plate wave excitation using bulk modes[3]. In this device, the silicon nitride membrane structure is fabricated on the bulk silicon substrate at first, which is then bonded to a glass substrate using anode bond technology. The microcavity below the silicon nitride membranes can be filled with fluid medium and microparticles. Two PZT plates are bonded to silicon and glass substrates, respectively, shown in Fig. 2. When a sinusoidal a-c signal with appropriate frequency and amplitude is applied onto the PZT plates, the bulk mode of the entire microdevice can be excited and is coupled to the membranes due to the energy coupling, driving the membranes vibrate as well. It is therefore to form a standing wave field in the microcavity. This ultrasound standing wave field can be utilized to separate the microparticles within the fluids. This microdevice achieves the on-chip microparticles separation and will do some basic work for the particle testing on the same chip in further.

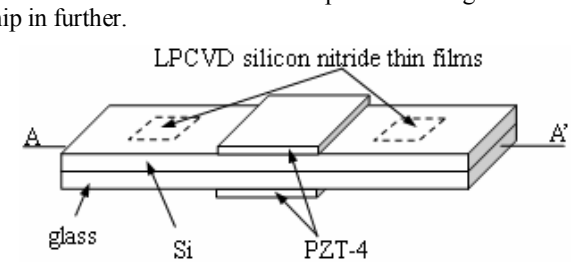


Figure 1. Schematic diagram of the proposed microdevice for on-chip ultrasonic separation of microparticles

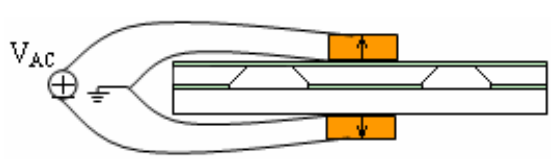


Figure 2. The A-A' cross-section of the proposed microdevice

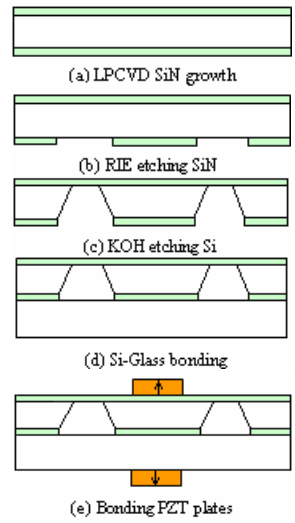


Figure 3. Fabrication process of the microdevice

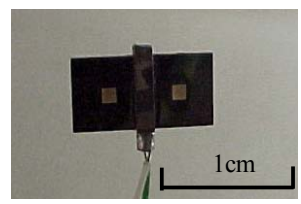


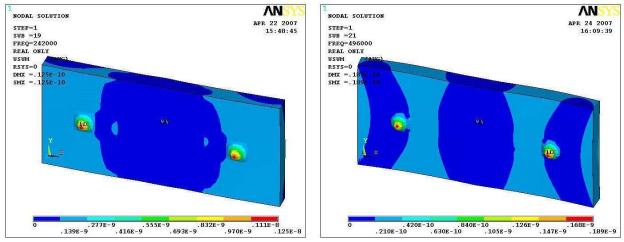
Figure 4. A sample of the fabricated microdevice

Fig. 3 shows the fabrication process of the proposed microdevice. First, the silicon nitride thin film of 1.3μm in thickness is grown on the (100) silicon substrate by using low stress LPCVD method, then the backside of the wafer is patterned and a window is formed by using RIE (reactive ion etching) to remove the patterned silicon nitride thin film. After that, the bulk silicon is etched by using KOH to form the silicon nitride membrane, and then the microfabricated device is bonded to a glass substrate. In the end, two PZT plates are bonded to the central parts of the bulk silicon and glass substrate, separately. Fig. 4 shows the fabricated microdevice. Compared with other methods, our microdevice is an on-chip separation one and the fabrication process is compatible to any CMOS-related process.

IV. Analysis of Proposed Microdevice

The ultrasonic microdevice is based on the principles of membrane vibration excited by bulk modes. When the PZT plates drive the bulk substrate into vibration, the silicon nitride membrane will also vibrate due to the energy coupling. In this paper, finite element method in ANSYS is used to study the microdevice.

With the modal analysis in ANSYS, we can obtain the bulk modes of the fabricated microdevmce, which As shown in Fig. 5, the silicon bulk substrate vibrates in longitudinal modes at 242 KHz and 496 KHz, and the two silicon nitride membranes vibrate as well due to the energy coupling.



(a) (b)

Figure 5. Bulk modes of ultrasonic separation microdevice: (a)f=242KHz; (b)f=496KHz

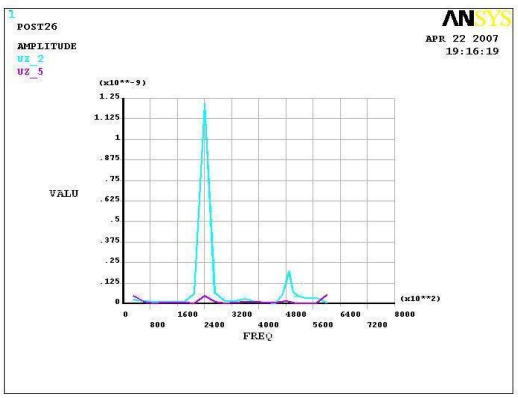


Figure 6. Vibration amplitudes of silicon nitride membrane and silicon bulk substrate in 2 bulk modes of the microdevice

The vibration amplitudes of silicon substrate and silicon nitride membranes at 2 different resonance frequencies are shown in Fig. 6, with the amplitude of membranes much larger than that of substrate. At resonance frequencies of 242 KHz and 496 KHz, the amplitude of membrane are 1.25nm and 0.19nm, respectively, and the vibration amplitude of the substrate is in the scale of pico meter, which can be neglected.

It is noted that this device can also vibrate in transverse modes or other modes, but the amplitudes of these modes at high frequencies are too small to be used in practice.

A ultrasonic stationary standing wave field is formed in the microcavity with the silicon nitride membrane vibrating. In order to calculate the acoustic radiation force, we must find out how much the acoustic/ultrasonic wave amplitude is, as described in Eq. (1) and eq. (2). We study the acoustic pressure caused by membrane vibration at first, and then find the relationship between the acoustic pressure and the ultrasonic wave amplitude.

When the membrane vibrates, the acoustic pressure at the membrane surface is

$$P = P_a e^{j\omega t} \tag{9}$$

where Pa is the pressure amplitude, ω is the circular frequency of the acoustic wave. The motion equation of the square membrane is

$$T(\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2})dxdy + Pdxdy = \sigma dxdy(\frac{\partial^2 \eta}{\partial^2 t}) \tag{10}$$

where T is the surface tension, σ is the surface density. The first item on the left is the total vertical force on the whole surface caused by surface tension, and the second item is related to the acoustic pressure. The item on the right of the equation is about the force acting on the membrane, which is derived from Newton's Second Law. By solving this equation, we obtain the displacement of the square membrane vibration that is expressed as

$$\eta(t,r) = \eta_a e^{j\omega t} = [A\cos kr - \frac{P_a}{2k^2 T}]e^{j\omega t} \tag{11}$$

Substituting the boundary condition of $\eta_{(r=b)}$=0 into above equation leads to the relationship between the membrane vibration amplitude and the acoustic pressure, which can be expressed as

$$P_a = \eta_a \frac{4\pi^2 f^2}{c^2}\sqrt{Ph}\,/(\frac{\cos kr}{\cos kb} - 1) \tag{12}$$

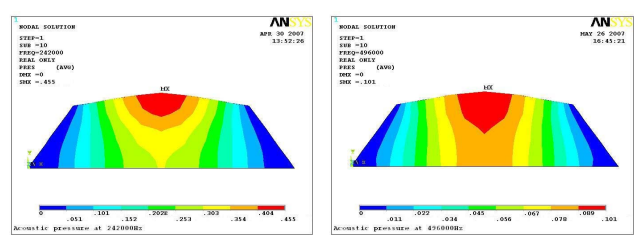
where η_a is the membrane vibration amplitude, b is the half length of square membrane.

According to Eq. (12), we can calculate the acoustic pressure caused by silicon nitride membrane vibration, which can also be verified by FEM in ANSYS. When the microdevice is working at 242 KHz or 496 KHz, the acoustic pressures in the microcavity are shown in Fig. 7. and the maximal acoustic pressure, which is in the center of the membranes, are 0.455Pa and 0.101Pa and listed in Table I. Calculated results from Eq. (12) are also listed in Table I. We can see that the results from two methods match well.

The relationship between acoustic pressure and acoustic wave amplitude is

$$A = \frac{P_a C}{4\pi f}\sqrt{\frac{r_0}{\rho_0}} \tag{13}$$

With this equation, the standing wave amplitude at different working frequency can be obtained and is shown in Table Ⅰ also. Thus, all parameters for calculating the acoustic radiation force on particles in Eq. (1) to (8) are determined.



(a) (b)

Figure 7. Acoustic pressure distribution of the standing wave field caused by membrane vibration: (a)f=242KHz; (b)f=496KHz

TABLE I. ACOUSTIC PRESSURES AND AMPLITUDES OF ULTRASONIC STANDING WAVE FIELD

Working frequency	Membrane vibration amplitude	Acoustic pressure (eq. 12)	Acoustic pressure (FEM)	Acoustic wave amplitude
242KHz	1.25nm	0.469 Pa	0.455Pa	30.35μm
496KHz	0.19nm	0.113Pa	0.101Pa	3.55μm

The particles can be separated from the medium when the separation force acting on them is positive, i.e. the acoustic radiation force is larger than the resultant force of viscous drag, gravity and buoyancy. The radiation force is principally determined by the frequency and wave amplitude. Furthermore, we can learn that whether the particle of a certain radius can be separated from the medium or not is principally determined by these two parameters. We analyze the separation force on a hemoleukocyte cell of 10μm in diameter suspended in blood plasma when our microdevice works on different working mode and find that the ultrasonic frequency and wave amplitude of the two working mode of our proposed microdevice can ensure that the separation force is positive, the hemoleukocyte cell can be separated from the medium effectively. On the basis of the results shown in Table Ⅰ, we calculate the relationship of the diameter of hemoleukocyte cells and the separation force acting on the beads for the given two bulk modes. The results of are shown in Fig. 8, the separation force increases with the increasing of particle diameter. When the hemoleukocyte cell diameter is in the range of several nanoemeter to 10μm, the radiation force is at the scale of several pico Newton.

The separation cavity of the microdevice can be filled with different mediums and particles. The forces acting on different particles when the device works at two working modes are shown in Table Ⅱ.

According to Table Ⅱ, when the device is working at 496 KHz, the separation forces acting on the micrococcus, epiphyte and virus in pure water and erythrocyte in blood plasma are below zero, they can not be separated from the medium. But when the device is working at 242 KHz, all of these particles can be separated effectively, because the wave amplitude at 242 KHz is large than that at 496 KHz.

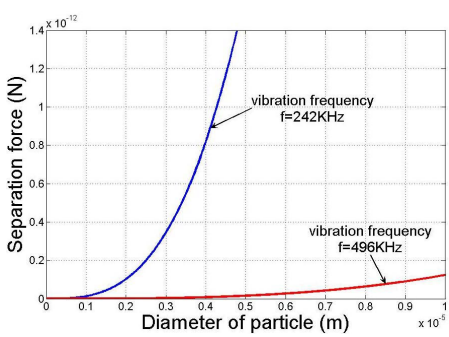


Figure 8. The relationship between the diameter of particles and the separation force

TABLE II. FORCES ON PARTICLES UNDER DIFFERENT VIBRATION FREQUENCY

Different particles suspended in mediums			Resonance frequency(KHz)			
			242		*496*	
Medium	Particle	Diameter (*μm*)	F_{ac} (pN)	F_{sp} (pN)	F_{ac} (pN)	F_{sp} (pN)
Blood plasma	Eryth-rocyte	8	13.1	5.82	1.54	-4.70
Blood plasma	Hemo-leukocyte	10	14.4	12.8	1.70	0.12
Pure water	Poly-styrene	4	2.34	2.17	0.28	0.11
Pure water	Micro-coccus	1	0.05	0.03	0.01	-0.02
Pure water	Epiphyte	6	11.1	4.88	1.31	-4.91
Pure water	Virus	0.1	4.49×10^{-4}	2.82×10^{-4}	0.53×10^{-4}	-1.15×10^{-4}
Milk	Milk fattiness	0.8	0.132	0.120	0.016	0.003

V. CONCLUSIONS

In this paper, we first study the acoustic radiation force on particles and then propose a new MEMS based microdevice for on-chip separation of microparticles. Working principle of the proposed microdevice and its numerical analysis using FEM in ANSYS have been made, and the separation forces acting on different particles when the microdevice is working are analyzed in detail. The results show that our proposed microdevice can be used to effectively separate the particles at the micron or even nano scale.

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